

Multiple Choice

1. **Answer:** D

Options: A) 1.5 R; B) 2 R; C) 2.5 R; D) 3.5 R

Note: Correct option: D - 3.5 R

2. **Answer:** D

Options: A) Ammonium hydroxide; B) Sulfuric acid; C) Acetic acid; D) Potassium hydrogen phthalate

Note: Correct option: D - Potassium hydrogen phthalate

3. **Answer:** A

Options: A) 3.74 T; B) 5.19 T; C) 6.08 T; D) 9.49 T

Note: Correct option: A - 3.74 T

4. **Answer:** D

Options: A) 10.8×10^{-5} ; B) 4.11×10^{-5} ; C) 3.43×10^{-5} ; D) 1.71×10^{-5}

Note: Correct option : D – 1.71×10^{-5}

4. **Answer:** A

Options: A) HCl; B) AgCl; C) CaCl₂; D) CCl₄

Note: Correct option: A - HCl

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Tl'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Oxide'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Blackbody radiation curves'.

Long Form Response

11. **Answer:** Both ^{35}Cl and ^{37}Cl have electric quadrupole moments.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $7.081 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.